



Student Name: _____

Instructor Name: _____

Q1. Select the correct answer for the following questions:

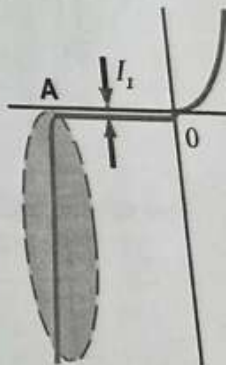
1. Due to the relative mobility factor in semi-conductors, GaAs compared to Si has relative mobility.
a. Higher. c. Lower. c. The same

2. In a half wave rectifier, if sine wave input is given as $200 \sin 300t$. Then, the average value of the output is approximately
a) 127.2 b) 67.453V c) 63.67V d) 76.987V

3. Using the ideal characteristic curve of normal diode, the resistance for $V_D < 0$ is considered as.....
a. 0. b. ∞ . c. Can not be determined.

4. The adding of Phosphorus (P) atom to a semi-conductor is creating element.
a. Electrically neutral b. Positively charged. b. Negatively charged. d. None of the above

5. After exceeding point A in this figure, the normal diode is and Zener diode is
a. Working, working.
b. Working, damaged.
c. Damaged, working.
d. Damaged, damaged.
e. Not mentioned.



6. The static resistance in a diode configuration is defined as
a. A point on the characteristic curve.
b. A tangent line at the Q-point.
c. A straight line between limits of operation.
d. None of the above.

7. The Zener diode in the breakdown region has.....

- a. Positive anode and negative cathode.
- b. Zero voltage.
- c. A knee voltage of the used semi-conductor.
- d. Open circuit equivalent.
- e. Negative anode and positive cathode.

8. The most popular application of normal diode (Not Zener or LED) in electronic circuit is to us

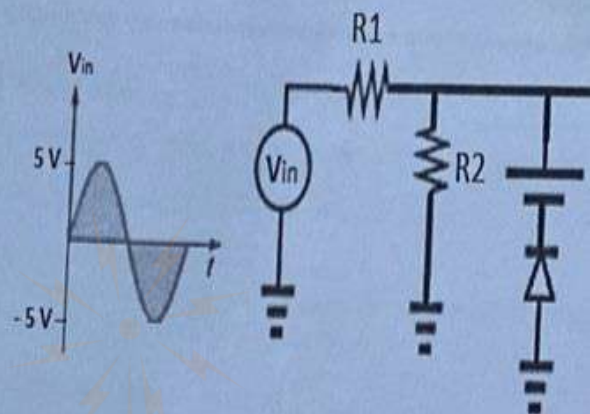
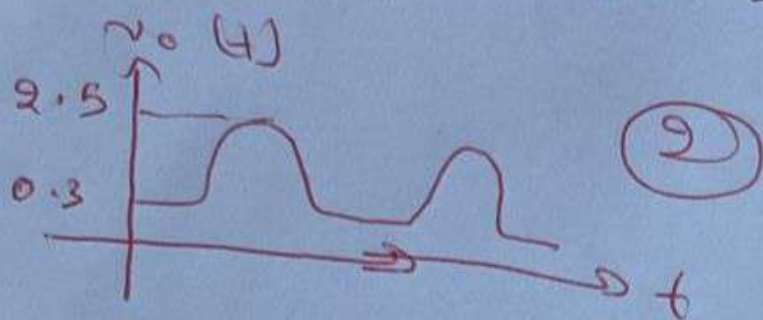
- a. Voltage regulator b. Switch c. Light Emitting c. None of them.

3. For the circuit shown below, assuming a Silicon simplified model of diode is used, if $V_2 = 1\text{ V}$ and $R_1 = R_2 = 1\text{ K}\Omega$, answer the following.

3.a. During the positive half cycle, what is the condition to have the diode in a forward bias?

$$V_{in} < V_2 \quad (2)$$

3.b. Draw the output voltage for the given input voltage.



Question 1 Answers:

1	2	3	4	5	6	7	8	9	10
a	c	b	a	c	a	e	b	b	b

(20)

10. For the given clamper circuit of an ideal diode, the capacitor voltage will

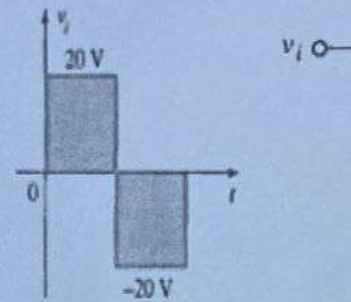
a. 35 V

b. 15 V

c. -5 V

d. -15 V

e. -35 V



Q2. For the given circuit below, D1, D3, and D4 are Silicon diodes and D2 is :
 $R_1 = 2\text{ K}\Omega$. Assume all diodes are simplified model diodes and then answer the following questions.

2.a. What is the status of the following diodes D1 and D3? Explain your answer.

D1: ON or OFF and why?

D1: on ①
 $V_0 > 0.7\text{ V}$ ①

D3: ON or OFF and why?

① off: parallel with V_{Ge} ①
 $V_{Ge} < V_{sin} \Rightarrow$

2.b. What is the current through D3?

$I_3 = 0\text{ mA}$ ①

2.c. What is the current through D1?

$-10 + 2I + 0.7 + 0.3 = 0$ ①
 $I = \frac{9}{2} = 4.5\text{ mA}$